

Labwork 3: Gray Image

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1 Implementation

An RGB image of size 6000×3375 was loaded and reshaped into $20,250,000 \times 3$ RGB values. Grayscale conversion was implemented using a CPU for-loop and a Numba CUDA kernel. Execution time was measured using `time.time()`.

The CPU execution time was 23.618 s and the GPU execution time was 0.0274 s, giving a speedup of approximately $862.16\times$.

2 Block Size Experiment

Different CUDA block sizes from 8 to 512 were tested. The execution time decreased from 0.0425 s to around 0.028 s. The fastest execution time was 0.02788 s with a block size of 256. Increasing the block size improves GPU performance at first, while the execution time remains nearly stable from block size 32 onward.

